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1. Introduction

1.1 Problem Background

1.1.1 Problem Background

Smart Nutrition leverages food image recognition and image processing to analyze meals and determine their nutritional values. By integrating computer vision with machine learning, the system accurately identifies food items and provides users with real-time insights into their dietary intake. Through automated meal analysis, Smart Nutrition enhances nutrition tracking, supports fitness goals, and promotes better dietary decision-making.

Traditional nutrition tracking methods often rely on manual food logging, which is both time-consuming and prone to errors. By automating this process, Smart Nutrition enables users to simply capture food images and instantly receive detailed nutritional insights such as calorie. Its ability to process unstructured food images makes it particularly beneficial for obesity patients, fitness enthusiasts, and healthcare applications, improving dietary management and overall health monitoring.

The importance of nutrition tracking is highlighted by the increasing prevalence of obesity and diet-related diseases. According to the World Health Organization (WHO), overweight and obesity rates have been rising worldwide, including Malaysia. A 2019 study revealed an alarming upward trend in obesity rates, increasing from 15.1% in 2011 to 17.7% in 2015, and reaching 19.9% in 2019. There are several factors which contribute to this trend, such as poor dietary habits and limited access to nutritious food. Poor dietary habits, which refers to the frequent consumption of high calories processed or fast food, along with sugary drinks. The modern fast-paced lifestyle further exacerbates the issue, as individuals often go for convenient but nutritionally imbalanced meals. Additionally, overweight individuals were more likely to be diagnosed with diabetes, hypertension, and high cholesterol, conditions often linked to poor dietary habits. The study also found that 94.7% of respondents did not consume the recommended daily servings of fruits and vegetables, highlighting the urgent need for better nutrition management. (Chong et al., 2023) Smart Nutrition plays a crucial role in addressing this

issue, particularly for healthcare professionals and individuals with obesity, by ensuring proper meal intake, preventing excessive calorie intake and promoting healthier eating habits.

1.1.2 Problem Statement

Our research highlights a significant lack of applications specifically designed to track the calorie content of Malaysian cuisine. While several food calorie tracking systems exist, such as the application "Lose It!", are primarily focused on Western food, leaving a gap in tracking local Malaysian dishes. Western cuisine typically features dairy, wheat, and mild herbs, with dishes like pasta, pizza, and tacos. In contrast, Malaysian cuisine is rich in spices, herbs, and coconut-based ingredients, offering a complex blend of sweet, spicy, sour, and savory flavors influenced by diverse culinary traditions. Several signature Malaysian dishes such as Chicken Curry, Nasi Lemak, and LoMian lack comprehensive nutritional data in existing tracking systems. This absence of an efficient and accessible Malaysia food tracking system makes it difficult for individuals to monitor their calorie intake accurately, leading to poor dietary habits and an increased risk of obesity-related health conditions.

Furthermore, without a reliable tracking solution, Malaysians remain unaware of the impact of their food choices, reducing the effectiveness of public health initiatives aimed at promoting balanced nutrition. To address this issue, Smart Nutrition leverages AI-powered food image recognition to automate meal analysis. By allowing users to capture an image of their meal, the system can automatically identify food items, estimate calorie content, and generate a detailed nutritional composition table of Malaysia cuisines. This approach eliminates the need for manual logging, improves dietary tracking accuracy, and provides real-time nutritional insights which help individuals reduce excessive calorie intake and improve adherence to balanced diets, ultimately supporting weight management and reducing obesity rates.

1.1.3 Question

- How does the application of Machine Learning, specifically Computer Vision, enhance the accuracy and efficiency of food recognition and calorie intake tracking in Malaysian cuisine?
- Among various deep learning algorithms (YOLOv8, InceptionV3, MobileNet, CNN), which offers the best performance in terms of accuracy and speed?
- Among traditional machine learning classifiers (SVM and KNN), which offers the best performance in the terms of accuracy and speed?
- What are the challenges when training different models for the food recognition system?

1.1. Objectives/Aims

- Analyse and compare different machine learning and deep learning algorithms in terms of efficiency and accuracy.
- Evaluate the performance of deep learning algorithms to identify the most accurate and efficient model for food recognition and calorie estimation.
- Evaluate the performance of traditional machine learning classifiers to identify the most accurate and efficient model for food recognition and calorie estimation.
- Identify and analyze the challenges faced when training different machine learning and deep learning models for food image recognition.

1.2. Motivation

The increasing prevalence of obesity and diet-related health issues in Malaysia underscores the urgent need for efficient nutritional tracking tools tailored specifically to Malaysian dishes. This project aims to provide accurate food recognition with calorie estimation tool which is essential for monitoring daily intake. The social impact of this solution is valuable, as it assists individuals in making informed dietary choices, reducing the risk of diet-related diseases such as diabetes and cardiovascular problems. Furthermore, the commercialization potential of this technology is promising. The system can be further integrated into health and fitness applications such as diet planner and telemedicine platforms. Therefore, this technology can attract a wide range of target users, including fitness enthusiasts, healthcare professionals and the general public who want to maintain a balanced diet but lack knowledge about calorie intake. Moreover, this project aligns with Malaysia's digital transformation goals by incorporating AI into health and fitness applications. The development of Malaysian local food recognition with calorie estimation tools can serve as a foundation for the future advancements in the healthcare sector.

2. Research Background

2.1. Background of the applications

Food image recognition and nutritional tracking have gained significant attention in recent years due to the rising prevalence of obesity and diet-related health conditions. Traditional calorie-tracking methods rely on manual input, which is often time-consuming and prone to human error. With advancements in machine learning and computer vision, food recognition applications have emerged as an innovative solution to automate dietary assessment. These applications leverage deep learning techniques to analyze food images.

Many studies have explored food recognition using deep learning and image processing techniques, demonstrating significant progress in food classification. However, existing research primarily relies on datasets such as Food-101, UEC-Food 100, and ChineseFoodNet, which focus on Western and East Asian cuisines, leaving Malaysian cuisine underrepresented. Due to the diverse and complex nature of Malaysian food, these models often struggle with accurate classification. While studies such as *Classification of Food Objects Using Deep Convolutional Neural Networks with Transfer Learning* and *Applying Image-Based Food-Recognition Systems on Dietary Assessment: A Systematic Review* have shown promising results in improving recognition accuracy, they are still limited by the lack of a comprehensive dataset for particular Malaysia food. This gap underscores the need for a specialized dataset and enhanced deep learning models tailored to accurately identify Malaysian food and support automated dietary assessment and calorie tracking. To address this, our project will focus on integrating AI-powered food recognition specifically designed for Malaysian cuisine to improve accuracy in food classification and calorie estimation.

Deep learning has revolutionized food image recognition by enabling automated meal analysis with high accuracy. Several studies have employed convolutional neural networks (CNNs) for food classification, with models such as Inception V3-based, ResNet, and MobileNet demonstrating strong performance in image-based food recognition. It achieved better performance than the traditional machine learning methods. For example, the FoodieCal system,

utilizing an Inception V3-based CNN model, has achieved an accuracy of 89.48% for food detection and calorie estimation. Similarly, *Food Image Classification and Data Extraction Using Convolutional Neural Network and Web Crawlers* reports a 97% accuracy for the Inception V3 model. In contrast, *Food Image Recognition Based on ResNet* examines the effectiveness of the ResNet algorithm, demonstrating that it outperforms Inception V3 in certain cases, achieving over 90% accuracy on the Food-101 dataset.

The authors in Kagaya et al.(2014) have applied CNN to the tasks of food detection and recognition through parameter optimization. They compared it with conventional methods using a support vector machine (SVM) with hand-crafted features. Their research covered 10 food classes, and the results showed that CNNs outperformed other conventional approaches, with a detection accuracy of 73.7%.

The author in Taskiran & Kahraman (2019) compares five different Convolutional Neural Network architectures including ResNet50, VGG16, VGG19, MobileNet and GoogleNet. The food dataset was used to evaluate the performance. The result showed that InceptionV3 obtained the highest Top-1 accuracy of 87.16%, followed by VGG16 with a Top-1 accuracy of 84.48, when using 70% as the training set and 30% as the testing set.

The authors in C et al. (2018) have applied the K-Nearest-Neighbour and Support Vector Machines to compare the performance and accuracy between these two models. Their research covers 5 Indian food classes. Their results show that the Support Vector Machine model performs better with a slightly higher detection accuracy compared to the K-Nearest-Neighbours model. For example, the food category, Chapathi with the SVM has achieved 75.43% accuracy while using the KNN only achieved 70.24% accuracy.

In Computer Vision and Image Processing, the paper titled “*You Only Look Once: Unified, Real-Time Object Detection*” by Redmon et al. (2015) introduces the YOLO algorithm, a pioneering approach in real-time object detection. While this foundational work does not specifically address food image recognition, it lays the groundwork for numerous subsequent applications in various domains, including food image recognition. Furthermore, Fitroh Romadhon et al. (2023) implemented the YOLO algorithm to identify food types and estimate their calories values based on data from FatSecret Indonesia. The paper provides a comprehensive analysis of YOLO’s performance in the context of food image detection and calorie estimation. It highlights that in the local testing environment, the YOLO’s performance reached 0.94 Precision, 0.90 Recall, and 0.91 F1-Score; while after their deployment on Hugging Face

Platform, the YOLO's performance drop significantly to 0.84 Precision, 0.32 Recall, and 0.41 F1-Score, due to the suboptimal CPU usage and reduced image quality during the upload process. The study titled *"Food Nutrition and Calories Analysis Using YOLO"* (Nivedhitha P et al., 2022) demonstrates that by training the YOLOv4 algorithm on a dataset comprising approximately 10,000 food images, the system achieved an accuracy of up to 92.37%. This performance surpasses that of earlier models, such as Convolutional Neural Networks (CNNs), highlighting YOLO enhanced capability in food image recognition.

2.2. Analysis of selected tool with any other relevant tools

Tools comparison	Google Colab	Jupyter Notebook	Streamlit	Gradio	Draw.io
Type of license and open source license	Free Tier: MIT-like Paid Tier: Proprietary (Pro/Pro+)	3-Clause BSD License	Open-source: Apache 2.0 License	Open-source: Apache 2.0 License	Open-source: Apache 2.0 License
Year founded	2017	2015	2018	2019	2000
Founding company	Google Research	Project Jupyter	Streamlit by Adrien Treuille, Thiago Teixeira and Amanda Kelly	Gradio Inc. (Acquired by Hugging Face in 2021)	JGraph Ltd.
License Pricing	- Free - Pro: \$9.99/mo - Pro+: \$49.99/mo	Free (open-source)	Free (open-source)	Free (open-source)	Free: Web app and desktop. Atlassian Cloud: \$1.70–\$0.07/user/month. Data Center: \$6,250–\$13,000/year. Server:

					\$10–\$450/year (team size-based).
Supported features	- GPU/TPU acceleration - Real-time collaboration - Cloud execution - Python/Markdown support	- Support over 40 programming language - Interactive code execution - Data visualization - Kernel support	- Fast web app building for Data Science - Built-in Widgets for user interaction - Data visualization	- Easy UI for ML models - Drag and drop UI - Supports image/text/audio/video inputs	- Real-time collaboration - Drag-and-drop diagram builder
Common applications	- Machine Learning - Deep Learning experiments - Educational notebooks	- Data Science - Research - Prototypes	- ML model deployment - Interactive dashboard	- ML model demos - Prototypes	- Software design diagrams - System architecture - Database modeling
Customer support	- Google forum - Community (StackOverflow, Reddit)	- Jupyter Community forum - GitHub issues	- Streamlit docs - GitHub issues	- GitHub issues - Hugging Face community forums	- GitHub issues - Online documentation

Limitations	- Limited ram, disk space and runtime (12 hrs for free, 24 hrs Pro) - Slower startup for large files - Requires internet	- Limited version control and poor support for testing/debugging. - Manual setup for dependencies - Slow with large dataset	- Not ideal for complex front-end UI - Not build for heavy backends	- Less control over UI styling - Not ideal for large-scale production apps	- Not as powerful as Figma for UI/UX wireframing
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2.3. Justify why the selected tool is suitable

Google Colab was chosen for developing our project because it provides free access to GPUs and TPUs, which significantly speeds up the model training process which is time consuming. Its cloud-based environment eliminates local setup issues and the real-time collaboration feature is useful for sharing the notebooks during group development. Thus, we can view the project progress and each other's work anytime.

Besides, Streamlit was chosen as the tool to build prototypes of this project because it is lightweight and faster development. It allows the creation of user-friendly interfaces with just a few lines of Python code. As the prototype of our project just requires some simple user interface that allows users to upload food images and receive real-time prediction, Streamlit is more suitable for creating the interactive web.

Moreover, the Draw.io tool was selected to draw the system flowchart as it is simple to use and supports drag-and-drop features ideal for drawing system design diagrams required for documentation.

3. Methodology

3.1. System Flowchart / Activity Diagram

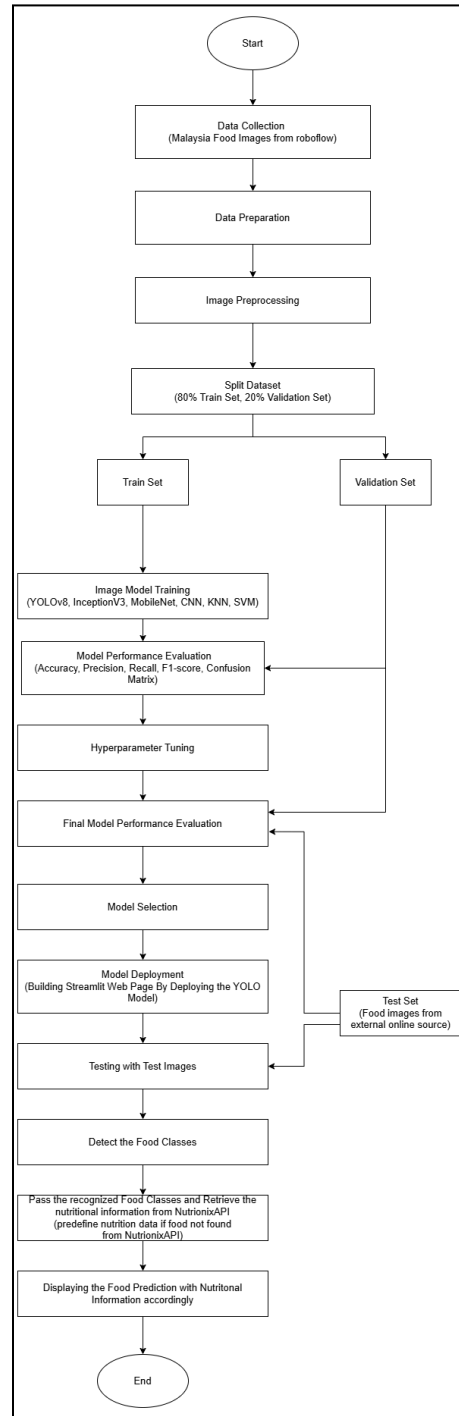


Figure 3. 1.1 Simple System Flowchart

Data Collection and Preprocessing:

Data Sources: Gather a variety of diverse and representative food images from publicly available repositories, such as those on Robotflow.

Data Preparation: For the dataset in Tensorflow Object Detection format along with `_annotations.csv`, the filename column of the dataset will be modified and updated with the correct directories to access the image correctly. The class label column will be encoded from string names to numeric index. For example, “Chicken Chop” is class 0, “Chicken Rice” is class 1, “Fish Soup” is class 2 and so on.

Image Preprocessing:

Algorithms	Image Preprocessing Techniques
Convolutional Neural Network (CNN), Support Vector Machine (SVM) and K-Nearest Neighbours (KNN)	• Convert images to grayscale • Resize the images to 224 x 224 pixels for a consistent input size across all images • The pixel values are normalized to [0, 1] range by dividing by 255.0 • CNN: Add channel dimension to each normalized 2D image (224 x 224), matching the input size requirement for CNN • SVM : Sharpen the image using the filter2D with Sharpening Kernel • KNN: Each 2D image (224 x 224) is flattened into a 1D vector
You Only Look Once Version 8 (YOLOv8)	According to the Ultralytics documentation, YOLOv8 automates several data preprocessing steps, including reading images, parsing annotation files, resizing inputs, normalizing pixel values, and performing data augmentation, all based on the configurations specified in the data.yaml file. In the case of our Robotflow dataset, preprocessing steps such as auto-orientation, resizing to 640x640, and applying augmentations like horizontal and vertical flipping have already been performed. Consequently, the data.yaml configuration has been

	appropriately set to reflect these preprocessing operations.
Inception-v3	• Rescale the image pixel values from [0, 255] to [0, 1] normalization • Resize the images to 224 x 224 pixels with 3 color channels (RGB), matching the input size requirement for Inception-v3
MobileNet	• Resize the images to 224 x 224 pixels • Normalize the image pixel value to [0, 1] • Normalize the bounding box using original images

Data Split: Divide the dataset into training and validation sets, typically in a ration of 80-20.

Model Selection and Configuration

Training: Train the CNN, SVM, KNN, YOLOv8, Inception-v3, and MobileNet algorithms on the training dataset with the selected parameters. During training, apply techniques such as custom layers, learning rate scheduling, early stopping, and cross-validation to ensure robust performance and avoid overfitting.

Model Performance Evaluation: Evaluate the performance of each model using metrics such as accuracy, precision, recall, and F1-score to assess classification performance. Additionally, for object detection models like YOLOv8, use metrics such as mean Average Precision (mAP) to measure how well the model detects and classifies objects within images.

Hyperparameter Tuning: Experiment with different hyperparameters to optimize the performance of each model. Key hyperparameters to tune include custom layers, learning rates, kernel sizes, dropout rates, and activation functions. Use different techniques to adapt to each algorithm to find the best hyperparameter configurations for each algorithm.

Final Model Performance Evaluation: After tuning the hyperparameters and training the models, evaluate the final performance of each model using a comprehensive set of metrics. For classification models, use accuracy, precision, recall, and F1-score to assess how well the model predicts the correct classes. For object detection models like YOLOv8, evaluate using Mean Average Precision (mAP), Intersection over Union (IoU), and precision-recall curves to determine how well the model detects and classifies objects within images.

Model Selection: Compare the models' performance in terms of accuracy, precision, recall and F1-score, ensuring the selected models meet the desired trade-offs between the performance.

Model Deployment:

Deployment: Deploy the trained YOLOv8-based food recognition model into the desired user-interface application, such as a Streamlit web service. This service will allow users to interact with the model in a simple, user-friendly interface, enabling real-time food image recognition.

Testing with test images: Users can upload food images to the Streamlit web service. The model will process the uploaded images, and the system will verify if the predicted food items match the corresponding food objects. The user will receive feedback on the predictions, including the identified food items among the 11 classes and confidence scores.

Gather the nutritional information: Once the food objects are recognized by the YOLOv8 model, the system will send the identified food items to the Nutritionix API to retrieve the corresponding nutritional information, such as calories, macronutrients, and vitamins. If the Malaysian local food items do not exist in the Nutritionix database, a predefined rule-based approach will be applied for nutritional information look-up, ensuring the system can still provide basic nutritional data based on local ingredients and food items.

3.2. Description of dataset

We sourced the data from Robotflow, a widely used platform for building computer vision applications. Roboflow offers tools to annotate, preprocess and export dataset in several formats which are an ideal option for our computer vision project. As our project topic focuses on Malaysia Local Food, thus we gathered the Malaysia Food images from various projects in Robotflow. The Local Food Computer Vision Project dataset which consists of Chicken Chop, Chicken Rice, Fish Soup, Kaya Toast, Nasi Lemak, Popiah, Porridge, Prata and Prawn Noodle are the main one of the main sources of our dataset. Additionally, we have incorporated data from the Kuih Lapis Computer Vision Project dataset and Wan Tan Mee Computer Vision Project dataset. To merge the various datasets, we fork the repositories in the Robotflow, a feature provided by Robotflow, resulting in a total of 3130 images dataset. This feature enables users to customize and integrate between several datasets that are needed. Robotflow allows the users to download in different formats prepared for image training. For example, the yaml file will be downloaded along with the images and labels annotations for dataset training in YOLO format; while the Tensorflow Object Detection format dataset contains a csv file consisting of annotation details in 8 columns such as the ymin ymax, xmin, xmax, image width and height, the filename and the label class of photo.

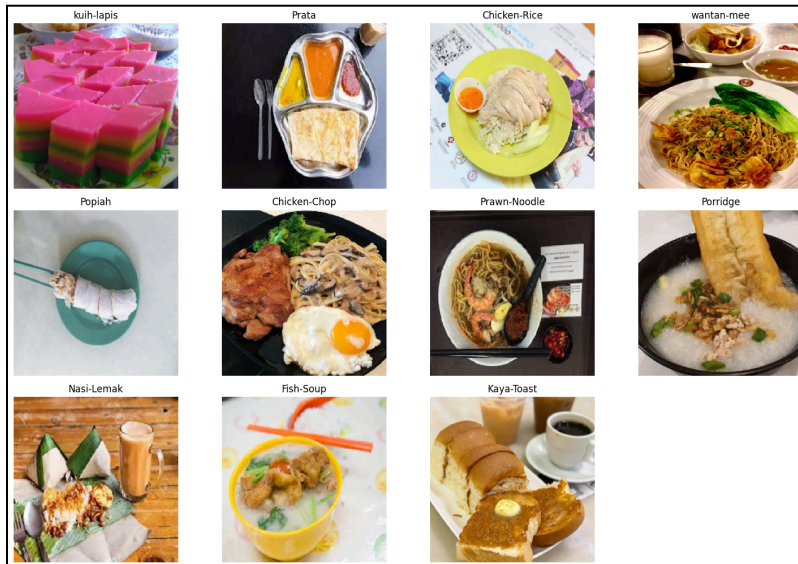


Figure 3.2.1 Dataset Images Classes

3.3. Description of algorithm(s)

After exploring and researching throughout scientific literature about food image recognition, we have chosen several algorithms to perform object detection and classification tasks efficiently. These algorithms are chosen based on the ability of handling images data, accuracy, efficiency and suitability of the algorithm.

1. You Only Look Once version 8 (YOLOv8)

YOLO (You Only Look Once) is a cutting-edge real-time object detection model introduced in 2015 by Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi in their influential paper "You Only Look Once: Unified, Real-Time Object Detection." (Redmon et al., 2015)

YOLOv8, the latest version, is chosen for this project due to its outstanding performance in real-time object detection, particularly in food image recognition. YOLOv8 enhances the original framework by predicting bounding boxes and class probabilities directly from full images in a single pass, utilizing an anchor-free approach. This design significantly boosts both detection speed and accuracy, making it particularly well-suited for food recognition systems. In the context of food recognition, YOLOv8 excels at handling the varied shapes and arrangements of food items. The model divides the image into an $N \times N$ grid, allowing it to detect multiple food items of different sizes simultaneously such as Chicken Rice, Fish Soup, or Prawn Noodles within the same frame, even when objects overlap or are partially occluded with unpredictable presentations. YOLOv8's backbone, built on an advanced convolutional neural network (CNN) architecture, captures features at multiple hierarchical scales, enabling the model to detect intricate food patterns and textures. YOLOv8 introduces the novel C2f (Cross-Stage Partial) module, which enhances its ability to detect objects of different sizes by combining high-level, abstract features with low-level, detailed information. This integration enables YOLOv8 to quickly and accurately classify food types, making it highly effective for real-time, multi-object detection in dynamic environments. Its speed and capacity to handle overlapping objects make YOLOv8 ideal for applications like dietary tracking, where fast processing and accurate identification are crucial.

To train the model, a custom dataset is used, where each image comes with label files that specify the class of the food and the bounding box coordinates. The training process is guided

using these annotations along with a YAML configuration file, which helps set up the training parameters for the YOLOv8 model.

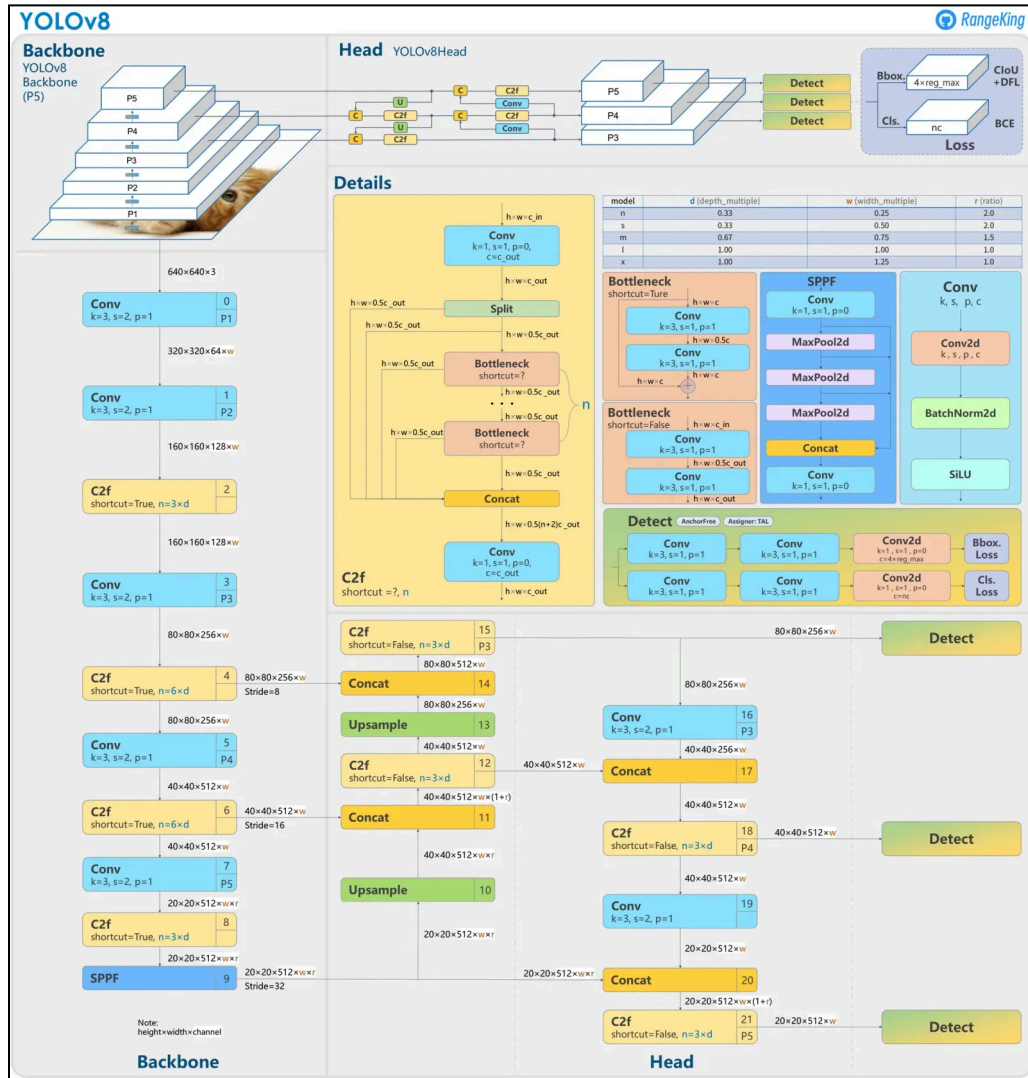


Figure 3.3.1 Architecture of YOLOv8

2. Inception-v3

Inception-v3 is a deep convolutional neural network (CNN) architecture developed by Google, designed for efficient and accurate image classification, including food image recognition projects. It excels at extracting complex features from images through its unique use of inception modules, which combine multiple convolutional filters of different sizes in parallel. It reduces computational complexity by breaking larger convolutions into smaller ones. For example, replacing a 5×5 convolution with two 3×3 convolutions. The architecture of Inception-v3 includes various layers such as convolutional layers, pooling layers, and inception modules that

help reduce computation while improving performance. InceptionV3 is commonly used as a feature extractor through transfer learning, where it is initially pretrained on a large dataset like ImageNet and later fine-tuned on a specific food image dataset. This method enables the model to accurately detect and classify a diverse range of food items, even under challenging conditions such as different lighting or presentation styles.

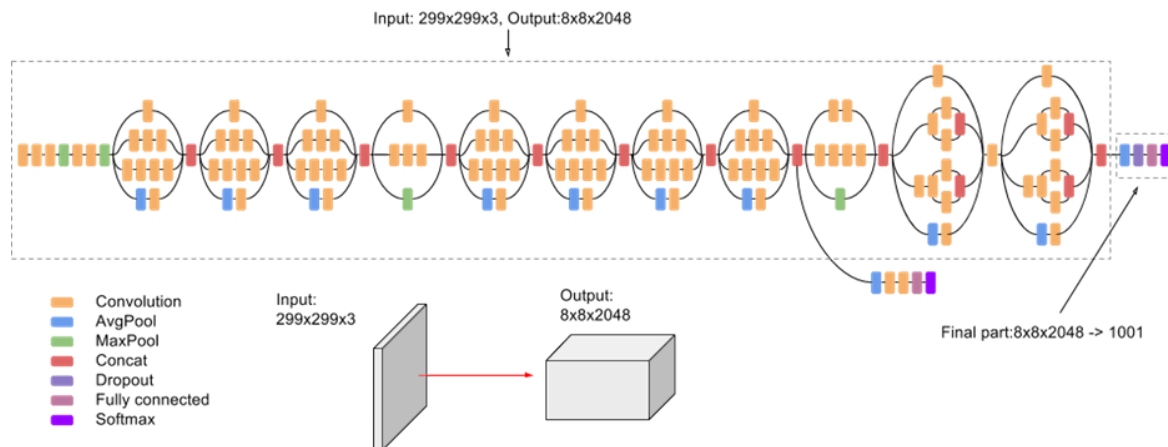


Figure 3.3.2 Structure of Inception-v3

3. MobileNet

MobileNet is a type of convolutional neural network which is designed to perform well on mobile devices and embedded systems. The MobileNet algorithm allows for quick and responsive detection that's still accurate enough for most daily food recognition tasks, especially when compared to the YOLOv8 algorithm. The MobileNet algorithm is implemented where the system will take an input image and process it. It will resize to the standard size of the images and normalise the pixels. The image input will then pass through a series of depthwise separable convolutional layers which consists of depthwise convolution and pointwise convolution. The depthwise convolution filters each input channel separately while the pointwise convolution combines these filters output across these channels. Each of the convolutional blocks is followed by batch normalisation and ReLU6 activation to introduce non-linearity and stabilize learning. Once the MobileNet algorithm has been trained, it can be easily deployed using tools like TensorFlow Lite or PyTorch Mobile. These tools convert the model into an optimized format that works well for mobile devices and embedded applications.

Due to its lower computational requirements, it is capable of handling various challenges that come with food image recognition tasks, such as the differences between angle and partial occlusion. The performance of MobileNet object detection in food image recognition tasks may not always match the performance of larger and more powerful models, however it is considered efficient and portable make it highly effective in everyday use cases, which is suitable to us as we are going to deploy food image recognition and nutritions tracking.

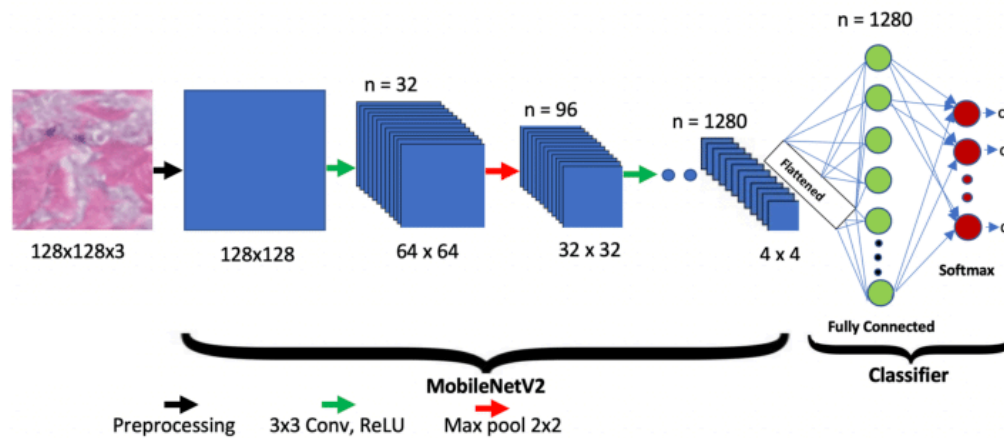


Figure 3.3.3 Structure of MobileNet

4. Convolutional Neural Network

Convolutional Neural Networks (CNNs) are a key type of neural network used for image classification. In deep learning, CNN models process food images by passing them through a sequence of layers, starting with convolutional layers that use filters (or kernels), especially 3x3 kernels to detect features such as edges, textures and patterns, which are crucial for distinguishing between visually similar food items. During convolution, multiple filters (such as 32 or 64 per Conv2D layer) slide over the input image, learning specific food characteristics like surface textures (e.g. crispy skin, smooth soup) or ingredient patterns (layers of Kaya Toast), combining its values with the image pixels to extract important features. An activation function, typically ReLU (Rectified Linear Unit) (Purwono et al., 2023), is applied after each convolutional operation to introduce non-linearity, allowing the model to learn complex patterns. In case, Batch Normalization can be applied after convolutions to stabilize learning and speed up convergence. Furthermore, the Dropout layers are strategically included to prevent overfitting by randomly disabling neurons during training. Pooling layers are then used to downsample feature maps using operations like max pooling or average pooling, reducing the spatial dimensions of the data

and the number of parameters as well as the computational cost while retaining important information. These layers enhance the network's robustness to variations such as food positioning and lighting conditions. Toward the end, fully connected layers consolidate the learned features and perform accurate food classification. The Softmax function is then used in the output layer to make the final classification through probability distribution across possible food categories. (Zhao et al., 2024)

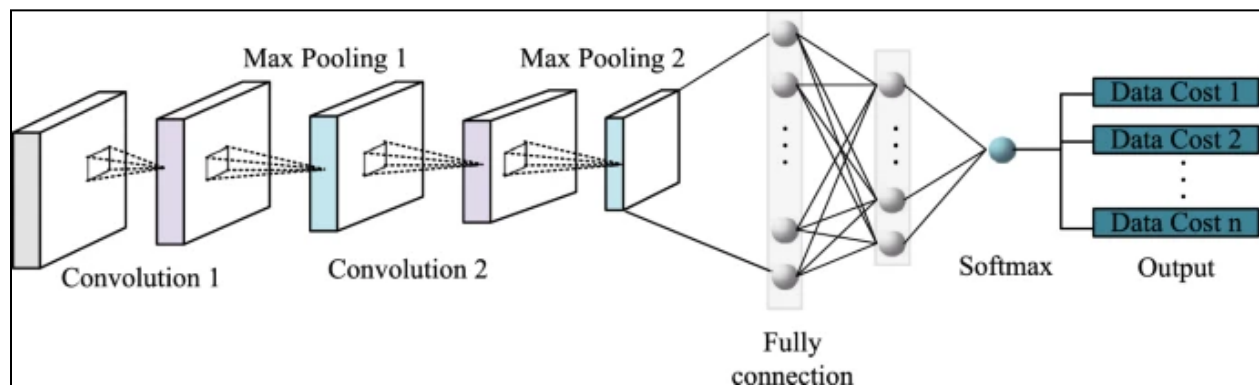


Figure 3.3.4 Structure of CNN

5. K-Nearest Neighbour (KNN)

K-Nearest Neighbors (KNN) is a traditional machine learning algorithm widely used for simple image classification tasks due to its simplicity and effectiveness. In this project, although KNN is not as efficient or powerful as modern deep learning models like Convolutional Neural Networks (CNN), it was still employed to evaluate the performance of classical machine learning techniques on food image recognition. By comparing the results of KNN with those of more advanced models such as Inception-v3, we can better understand the differences in performance between traditional machine learning and deep learning approaches for this task.

Additionally, it serves as a basis for comparing the performance of SVM, which is also employed in this project. KNN classifies a new sample based on the majority label among its nearest neighbors in the feature space. In our cases, these features are pixel values in image matrix (height x width) KNN algorithm classifies new unknown data points by finding the most common class among the k-closest examples. In order to apply KNN classification, we need to define distance metric or similarity function. In this case, Euclidean distance will be used to compare the images for similarity.

Euclidean distance:

$$d(\mathbf{p}, \mathbf{q}) = \sqrt{\sum_{i=1}^N (q_i - p_i)^2}$$

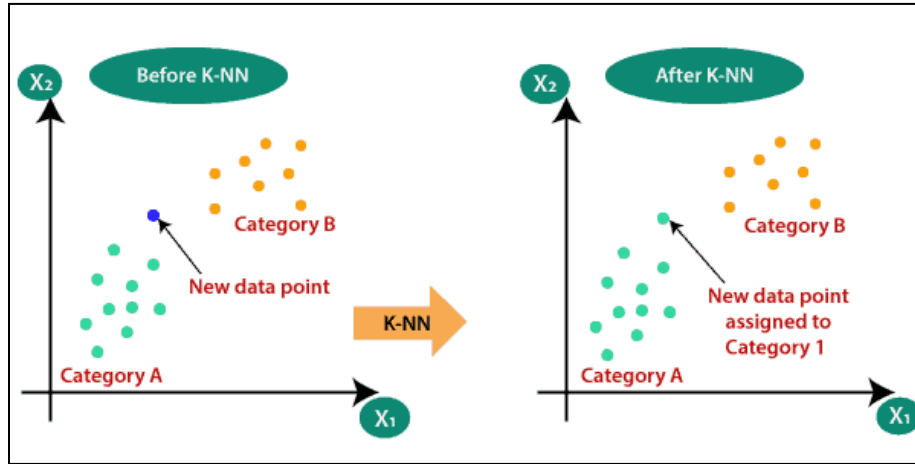


Figure 3.3.5 KNN Algorithm Training Process

6. Support Vector Machine

Support Vector Machine (SVM) was included as a baseline algorithm to provide a comparative understanding between traditional supervised machine learning approaches and deep learning algorithms. Although it is not typically flavoured in object detection tasks, SVM remains highly relevant for pure classification problems, particularly when isolated images of food. It is able to excel in high-dimensional spaces and particularly effective for multi-class classification tasks when it is paired appropriately with HOG embedding. SVM model for image classification is primarily applied after feature extraction of the particular images. Feature extraction technique is needed to be employed instead of just feeding raw pixels value which may lead to poor performances. The images will then transform into a structured features vector using HOG descriptor. The HOG will then capture edge directions as well as textual patterns to determine and differentiate between food categories. SVM works by constructing an optimal hyperplane that separates classes with maximum margin in feature space. Its resilience to overfitting and flexibility through kernel functions such as the radial basis function (rbf) kernel, polynomial kernel or linear kernel allows it to be suitable for tasks involving well-separated food categories. Radial basis function is used to capture and handle non-linear decision boundaries,

which may occurs more often in complicated food image, the polynomial kernel to capture interaction between the features while the linear kernel is implement for simpler or easier class separation scenarios. SVM was usually used to classify regions of interest after feature extraction, assisting in evaluation of how traditional models performed on image data.

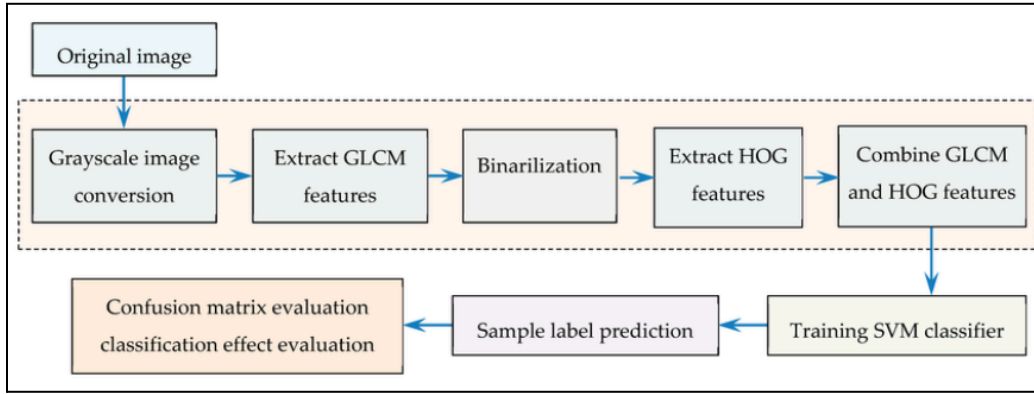


Figure 3.3.6 SVM Algorithm Training Process

3.4. Proposed test plan/hypothesis

Test Plan:

The performance of the models will be evaluated using performance metrics such as Accuracy, Precision, Recall, F1-score, Confusion Matrix and mean Average Precision (mAP).

Accuracy : It is the simplest matrix to evaluate the performance of the model. It is calculated as the number of correct predictions divided by the total observations in validation data, and multiplied by 100, where True Positives (TP) are the number of positive observations that are correctly classified, True Negatives (TN) are the number of negative observations that are classified as negative, False Positives (FP) are the number of negative observation that are falsely classified as positive and False Negatives (FN) are the number of positive observations that are falsely classified as negative.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

Figure 3.4.1 Accuracy formula

Precision : The number of observations are correct among the positively classified observations. In the context of food image recognition, if the model predicts 10 images as 'Chicken Rice' and 8 are correct, precision = 0.8. It is important when minimizing the false classification.

$$Precision = \frac{TP}{TP + FP}$$

Figure 3.4.2 Precision formula

Recall : It determines the proportion of actual positive cases that were correctly predicted. It measures the sensitivity of the model. High recall means very few real items are missed (low false negatives), the model successfully detects most food items.

$$Recall = \frac{TP}{TP + FN}$$

Figure 3.4.3 Recall formula

F1 Score : It is a metric utilized to evaluate the performance of classification models, especially when there are imbalanced classes. It is the harmonic mean of precision and recall. The range of F1-score is between 0 and 1. The closer to 1, the better the model. The larger the value of precision and recall scores deviate from each other, the lower the F1-score will be. F1-score is crucial for classification tasks as precision and recall are equally important.

$$F1score = 2 \times \frac{(Recall \times Precision)}{(Recall + Precision)}$$

Figure 3.4.4 F1-Score formula

mean Average Precision (mAP) : It is the average precision across all classes in the dataset. YOLO models use it to evaluate how well they classify objects. Higher mAP indicates the model is better at finding the object and correctly identifying food.

Test Objective:

To measure the classification accuracy in real-world cases, the food image recognition system will be tested by feeding it food images from a test set which consists of food images from external online sources.

Hypothesis:

Deep learning models (YOLO, InceptionV3, CNN and MobileNet) will achieve significantly higher classification and detection accuracy compared to traditional machine learning models (SVM, KNN) when applied to Malaysian Local Food Recognition, with Top-1 accuracy exceeding 85% for at least one deep learning model. YOLO is expected to demonstrate a mean Average Precision (mAP) above 80% for multi-food item detection, while Inception V3 and MobileNet are expected to achieve competitive classification accuracy comparable to larger. Traditional models (SVM, KNN) are expected to perform lower due to limited feature extraction capabilities on complex Malaysian Food images.

4. Result

4.1. Results

Table 4.1 Model Accuracy Comparison

Model	YOLOv8	Inception-v3	MobileNet	CNN	KNN	SVM
Accuracy Score (%)	97.5	97.2	96.06	68	53	60
Precision (%)	96.4	97	96.25	71	52	48.46
Recall (%)	96.1	97	96.06	68	53	47.59
F1 Score (%)	96.2	97	96.07	68	52	47.68

Table 4.2 YOLOv8 Model Evaluation Metrics After Hyperparameter Tuning

Model	Precision	Recall	mAP50	mAP50-95
YOLOv8	96.4	96.1	97.5	78.7

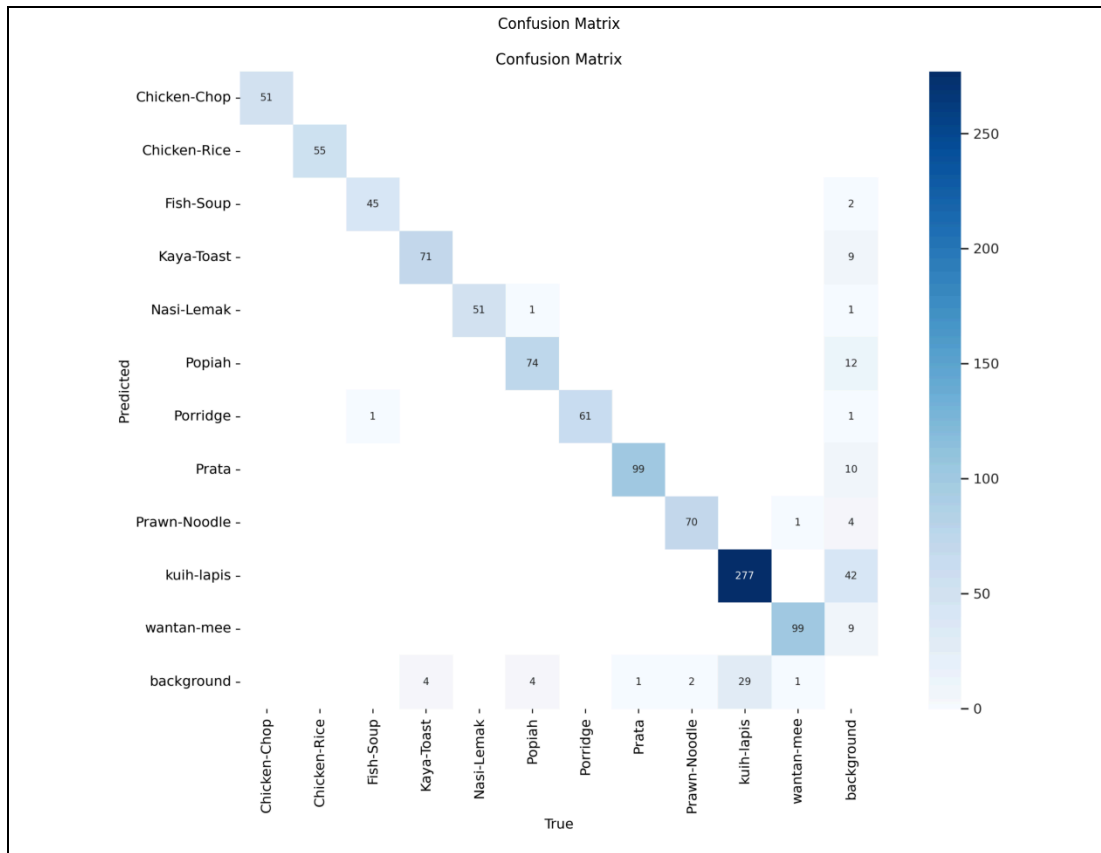


Figure 4.1.0 Confusion Matrix of YOLOv8 Model

Figure 4.1.0 demonstrates the confusion matrix of the YOLOv8 model after hyperparameter tuning. The YOLOv8 model performs high accuracy for some classes such as “Kuih Lapis”, “Prata” and “Wan Tan Mee”. For example, the model performs best at “Kuih Lapis” since it correctly predicts 277 instances, and Prata with 95 correct predictions. The “background” class in the confusion matrix represents instances where the model made a prediction that did not correspond to any actual object in the ground truth annotations, or where an actual object was not detected by the model. The model also might think the food images are background or vice-versa. For example, 71 actual Kaya Toast objects were correctly predicted, but 9 were misclassified as background. Moreover, a larger number of “Kuih Lapis” were correctly predicted but a significant number of 42 were misclassified as “background”. This could indicate the issues with detecting smaller or occluded “Kuih Lapis” objects. Certain food categories, such as “Popiah”, “Nasi Lemak”, and “Prawn Noodles”, show occasional misclassifications with other classes, but these errors are relatively infrequent compared to the correct predictions. For the

overall visualization of the confusion matrix of YOLOv8, the model is quite good but confusions are minimal and mainly between similar looking foods or background noise.

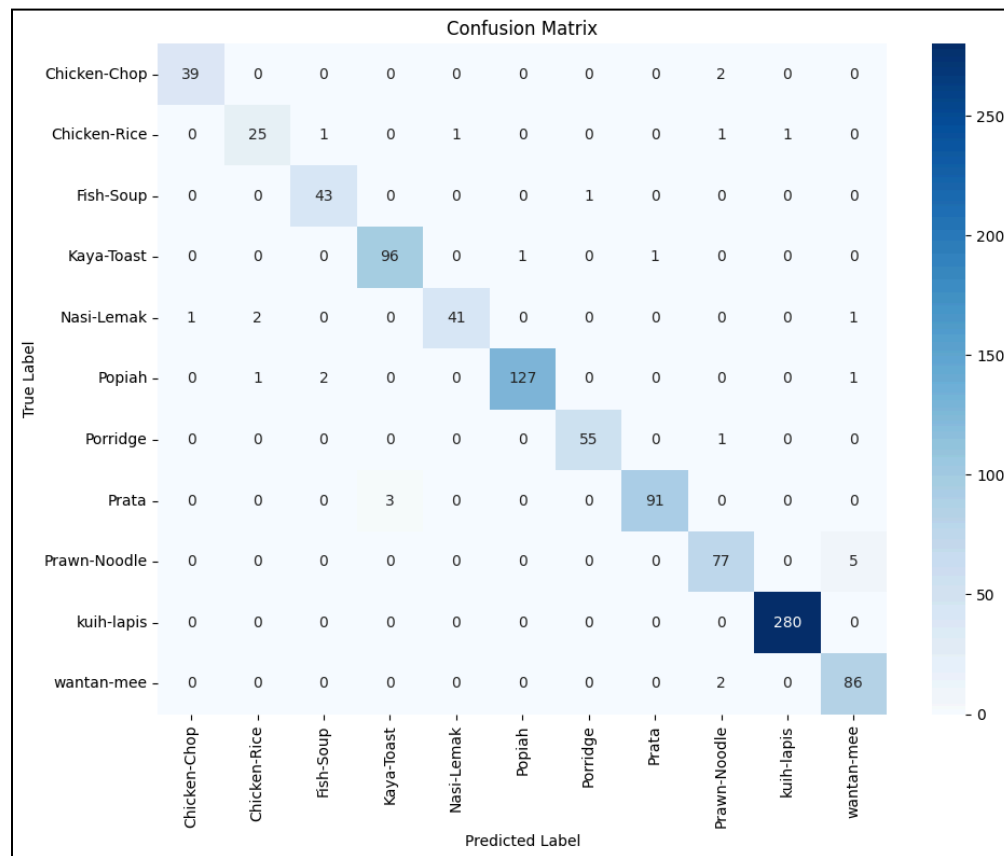


Figure 4.1.1 Confusion Matrix of Inception-v3

Figure 4.1.1 displays the confusion matrix of the Inception-v3 model. The confusion matrix visualizes the classification performance of the model across the 11 food classes. The result demonstrates high classification accuracy across most food categories, aligning with the reported accuracy of 97%. Specifically, the model performs explicitly well for several classes such as “kuih-lapis”, “Fish-Soup”, “Porridge”, and “Prata”. For instance, the model correctly classifies 91 out of 94 instances of the “Prata” class, indicating high accuracy. There are a few misclassification observed, such as “Prata” being confused with “Kaya-Toast”, “Prawn-Noodle” being confused with “wantan-mee” and “Nasi-Lemak” being confused with “Chicken-Chop”, “Chicken-Rice” and “wantan-mee”. The misclassification between “Prata” and “Kaya-Toast” are due to visual similarity, as both have a flaky and golden-brown appearance.

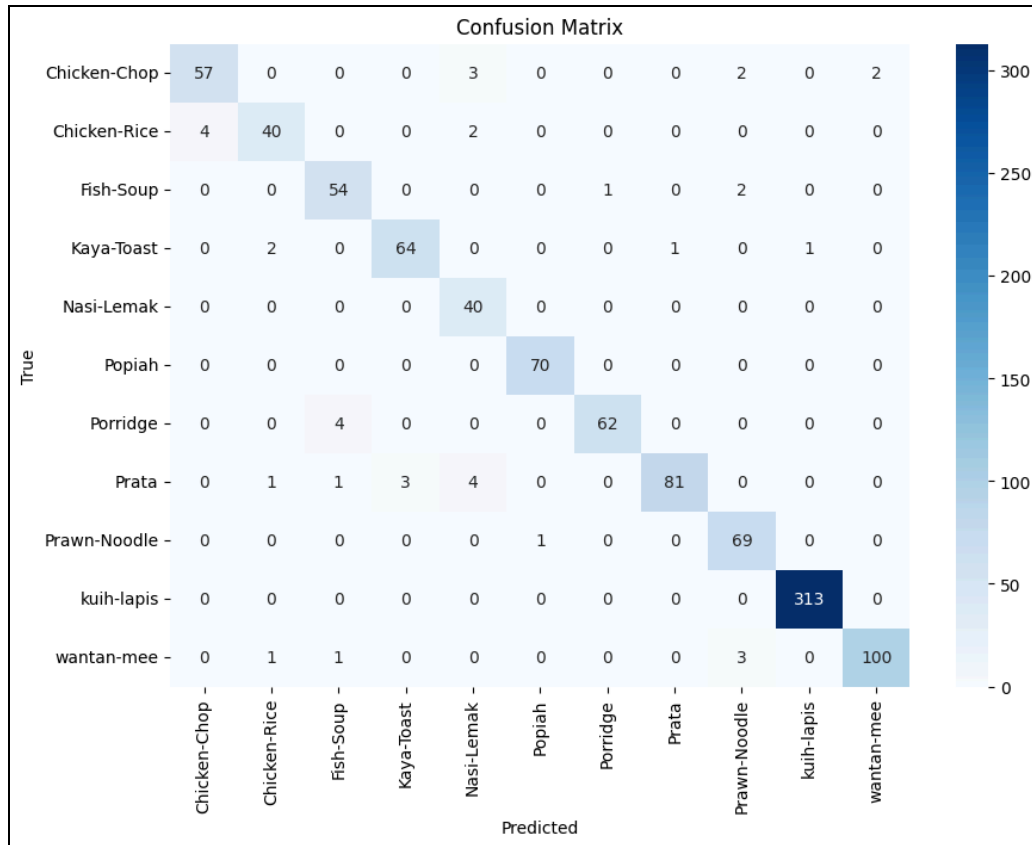


Figure 4.1.2 Confusion Matrix of MobileNet

Figure 4.1.2 shows the confusion matrix of the MobileNet model in food image recognition. The row represents the true labels while the columns represent the predicted label. The higher value appears diagonally, indicating that the model has accurate and correct predictions, whereas it means the image is misclassifications. Based on the confusion matrix diagram above, the “kuih-lapis” can be said to be perfectly classified with 313 correct predictions and 0 misclassification. Besides “kuih-lapis”, “Nasi-Lemak” and “Popiah” have also performed explicitly well which includes 40 instances and 70 instances without misclassification. The food class, “Prawn-Noodle” also reaches high accuracy with only 1 instance misclassified as “Popiah” over 70 instances which means it has a great performance in classifying this food class. Furthermore, “wantan-mee” class has correctly identified 100 out of 105 instances, demonstrating a high level accuracy for this category, although there is still some misclassification. Other classes, such as the “Porridge”, “Kaya-Toast”, “Fish-Soup” and etc have also performed well in the MobileNet model although there are still minor instances that have been classified wrongly. Overall, mostly food category predictions are tightly concentrated along

the diagonal, indicating a consistent and strong classifications among all 11 food categories, even though there are still some minor classification mistakes.

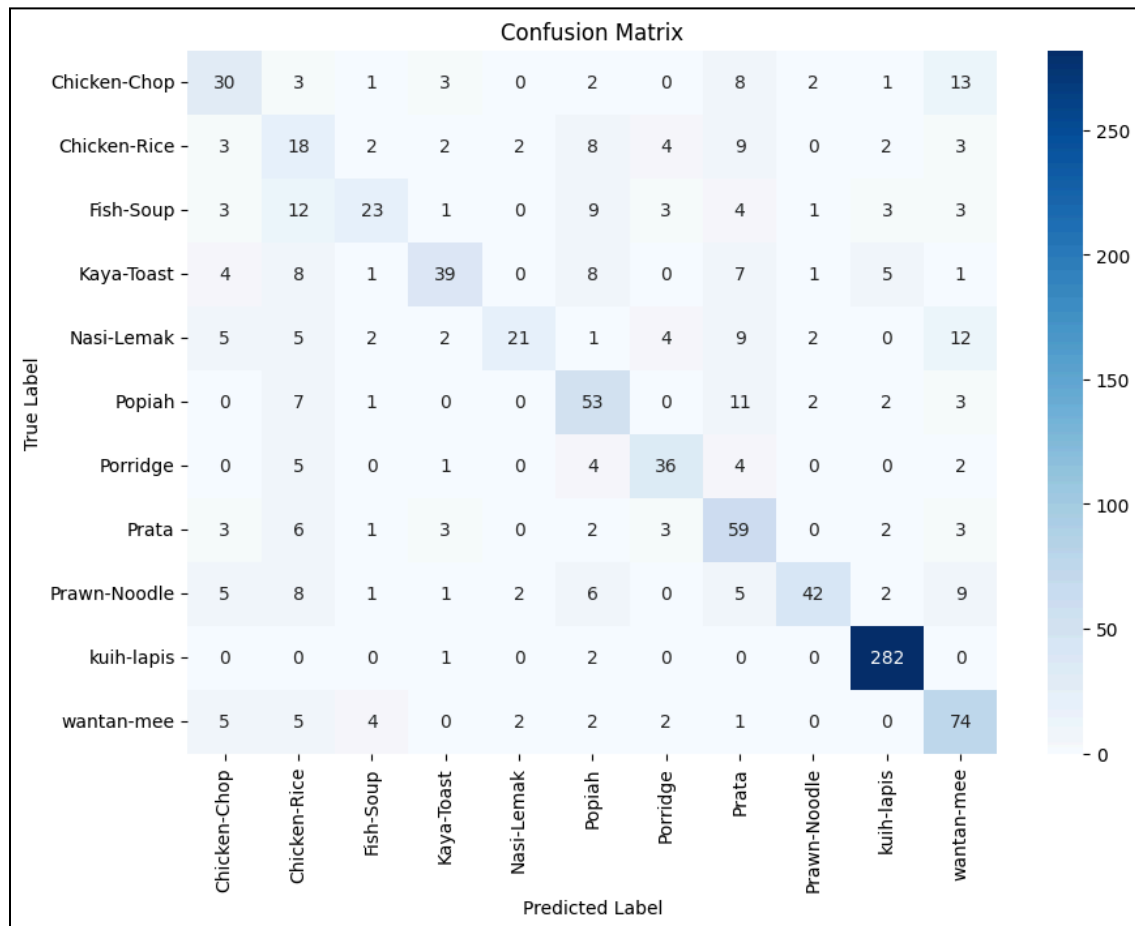


Figure 4.1.3 Confusion Matrix of CNN

Figure 4.1.3 shows the confusion matrix of CNN model. Based on the plot, the best performance is “Kuih Lapis” with 282 correct predictions with no misclassification as other classes, though 2 “Popiah” instances were mispredicted as “Kuih Lapis”. The “Prata” reaches 59 correct predictions with minor misclassifications such as 3 as “Chicken Chop”, 6 as “Chicken Rice”, 1 as “Fish Soup” and so on. The notable confusion is between the “Chicken Chop” and “Chicken Rice”. 3 “Chicken Chop” instances were predicted as “Chicken Rice” and 3 “Chicken Rice” instances as “Chicken Chop”. This indicates visual similarities as both are chicken-based dishes. Furthermore, 11 “Popiah” instances are predicted as “Prata”, suggesting that the overlap appears in similar textures or colors. These patterns suggest that CNN may be relying on certain visual features that are common across these classes, rather than distinguishing features.

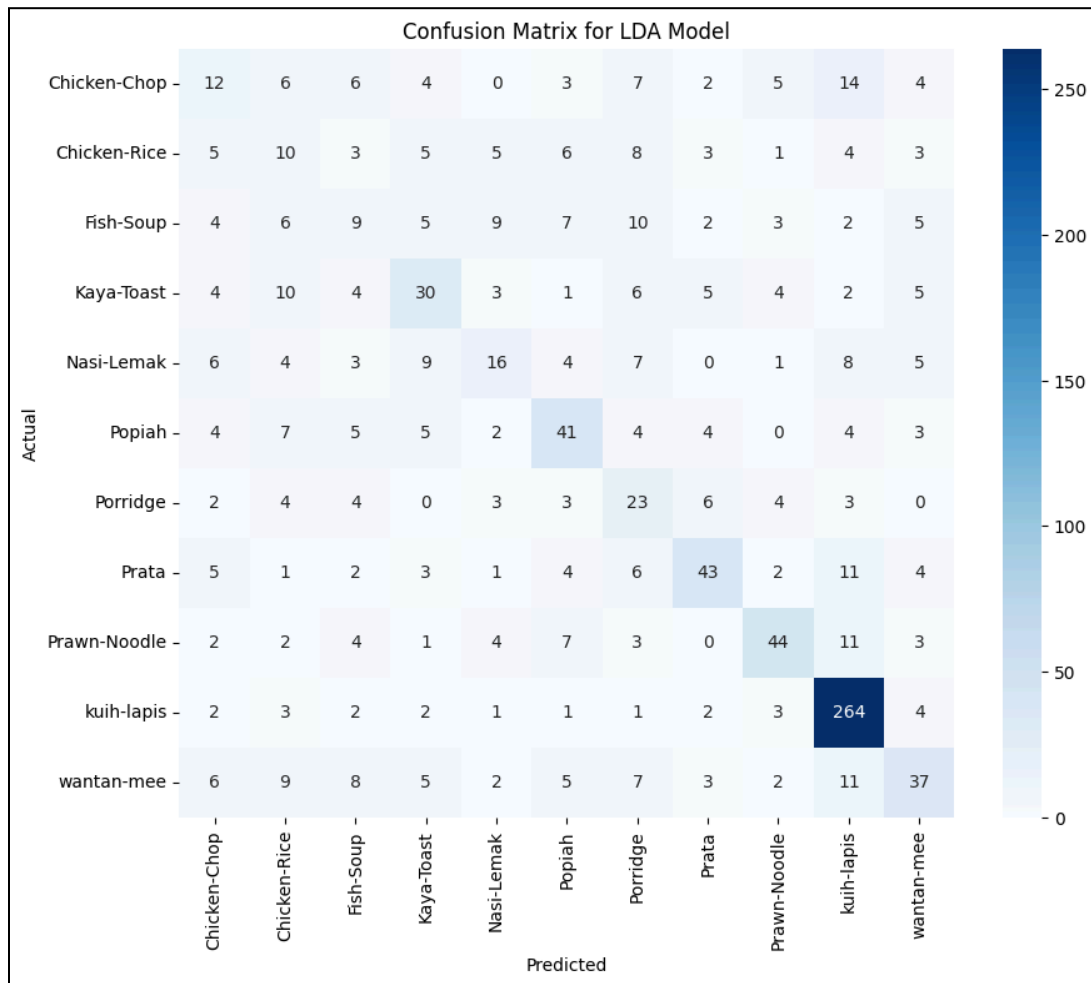


Figure 4.1.4 Confusion Matrix of LDA-based KNN model

Figure 4.1.4 displays the confusion matrix for the LDA-based KNN model. The result shows the weakest classification performance among 6 models. Despite its simplicity, it struggles with many categories, including some that are typically easier to classify, such as “kuih-lapis” and “Popiah”. It often struggles to distinguish between different dishes, even when they have clearly distinct features. Foods like “Chicken-Rice” exhibit particularly poor results, with only 10 out of 53 instances correctly classified, and the remaining misclassified across multiple classes, including 8 as “Porridge”, 6 as “Popiah”, and 5 each as “Chicken-Chop”, “Kaya-Toast”, and “Nasi-Lemak”. Even “kuih-lapis”, which has a distinct layered appearance, has 16 out of 280 instances misclassified, including 4 as “wantan-mee” and 3 as “Prawn-Noodle”. The possible reason for these misclassifications is the model’s limited ability to capture fine-grained distinctions. LDA’s linear dimensionality reduction oversimplifies the complex visual features of

these dishes, and KNN’s distance-based classification struggles to differentiate between overlapping feature representations, leading to frequent errors across classes.

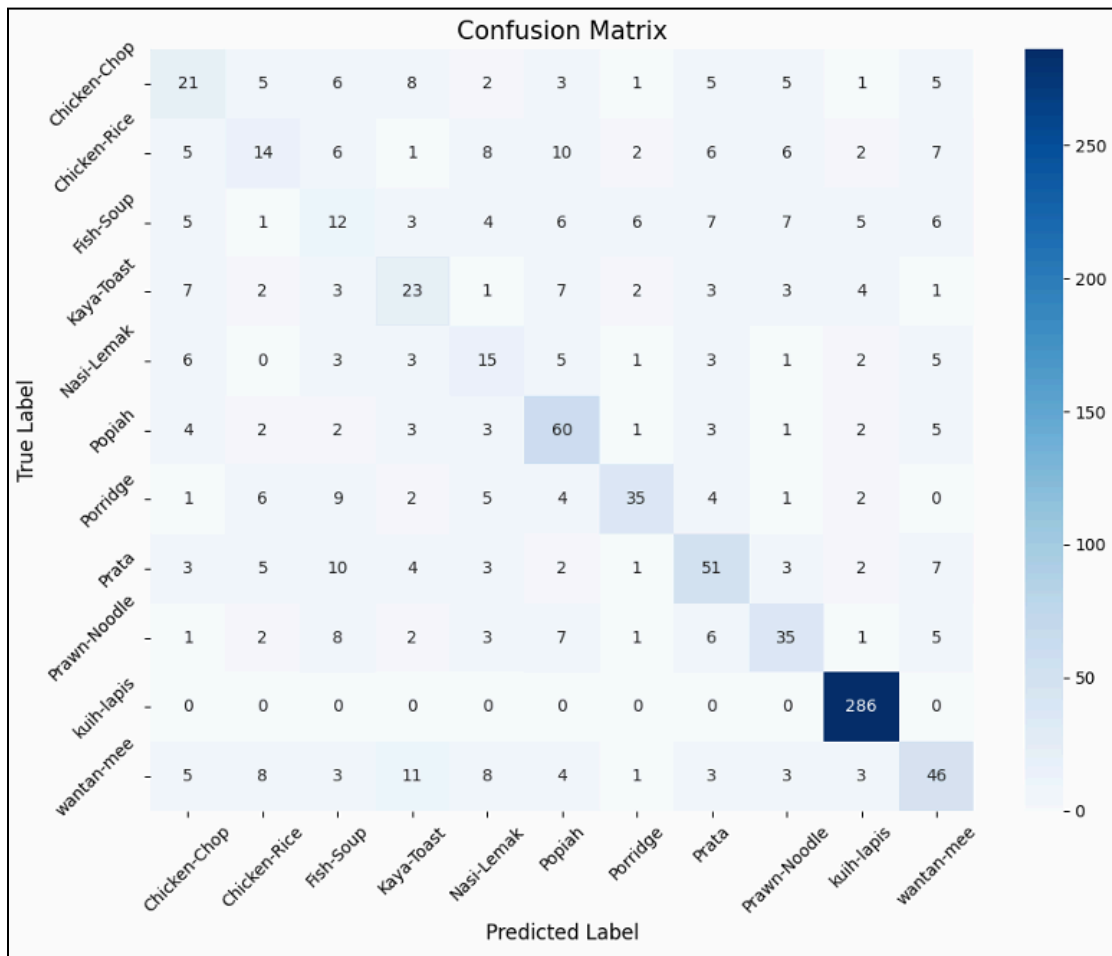


Figure 4.1.5 Confusion Matrix of Support Vector Machine (SVM)

Figure 4.1.5 represents the confusion matrix of the Support Vector Machine model in food image recognition. The row represents the true labels while the columns represent the predicted label. The higher value appears diagonally, indicating that the model has accurate and correct predictions, whereas it means the image is misclassifications. As we can see from the diagram, the SVM model performs extremely well in classifying the “kuih-lapis” class without a single error. This indicates that the SVM model learns well with this class. Besides the “kuih-lapis” class, the “Popiah ” class also displays a strong performance with 60 correct instances predictions and minor confusions, followed by the “Prata” class with correctly identified 51 out of 86 instances. On the other hand, “Fish-Soup”, “Chicken-Rice” and “Chicken Chop”

experience higher level of confusion as the model can only determine 21 instance of "Chicken-Chop" correctly with the total of 62 instance, 14 instance of "Chicken-Rice" accurately and 12 instance are identified correctly with the total of 62 instance of support. For instance, "Chicken-Chop" and "Chicken-Rice" are often misclassified with each other and with other food items, possibly because of their visual similarities. Overall, the SVM model shows displays strong performance in certain class, but it struggle with most of the categories which may the appearance may be more similar. Therefore, SVM is not suitable to be used in the food image recognition with calories estimation application as it cannot completely and accurately identify the 11 food categories.

4.2. Discussion/Interpretation

The proposed hypothesis stated that deep learning models (YOLO, InceptionV3, CNN, MobileNet) would achieve significantly higher classification and detection accuracy compared to traditional machine learning models (SVM, KNN) when applied to Malaysian local food recognition, with at least one deep learning model exceeding 85% Top-1 accuracy.

The experimental results supported this hypothesis. YOLOv8 and MobileNet achieved 97.5% accuracy, while Inception V3 achieved 97.2% accuracy, all surpassing the 85% threshold. Although Convolutional Neural Networks (CNN) have proven to be powerful tools in image classification tasks, in this study, the basic CNN architecture demonstrated lower performance compared to more advanced deep learning models such as YOLOv8, Inception-v3 and MobileNet. This results highlights that while CNNs are effective, deeper architectures are better suited for handling complexity of Malaysian local food images. In contrast, traditional machine learning models such as SVM and KNN achieved significantly lower accuracies of 60% and 53%, respectively. Therefore, the hypothesis is confirmed. Deep learning models clearly outperform traditional machine learning models for Malaysian Local Food Recognition tasks.

Based on Table 4.1, the F1-scores of three leading deep learning models show that Inception-v3 achieved the best score at 97%, while YOLOv8 and MobileNet scored 96.02% and 96.07%, respectively. Inception-v3 demonstrated the best overall balance between precision and recall, making it the most balanced model. However, it primarily focuses on single-label classification. In contrast, YOLOv8, despite having a slightly lower F1-score, offers the additional advantage of multi-food object detection within a single image. This feature is particularly valuable for real-world Malaysian food recognition, where multiple dishes often appear together. Given the focus of our project on calorie tracking, which requires multi-item detection, YOLOv8 would be the more suitable choice for our application.

5. Discussion and Conclusion

5.1. Achievements

Our project, the food image recognition and calories estimation has successfully achieved the goals that have been stated previously. We have explored, analyzed and compared several types of machine learning and deep learning algorithms to determine the best model to perform food image recognition and calories estimation accurately and efficiently. Models including YOLOv8, Inception-v3, MobileNet, CNN, KNN and SVM were evaluated based on key performance metrics, such as accuracy, precision, and F1 score.

In the aspects of deep learning algorithms, YOLOv8 offered the best performance compared to others. After constructing experiments on a diverse and well annotated food image dataset, YOLOv8 consistently demonstrated the best balance between accuracy score and speed. Therefore, it is the most suitable algorithm to perform food image recognition and calories estimation. YOLOv8 combines a single-stage detection approach with improved feature extraction and multi-scale prediction capabilities, allowing YOLOv8 to not only accurately classify food items but also localize them within images. Moreover, Inception-v3 also achieved high accuracy(97.2%), but it is computationally heavier and slower, making it less efficient for deployment in scenarios that require speed.

Among the traditional machine learning classifiers such as SVM and KNN, SVM outperformed KNN in the term of accuracy, with a slight difference of 7% higher accuracy for SVM. This can be explained by SVM's ability to find an optimal decision boundary and its robustness to noise and high-dimensional data compared to KNN. KNN relies on distance-based classification, which can struggle with high-dimensional data due to the curse of dimensionality, leading to less accurate classifications. However, it is important to note that, despite SVM's better performance over KNN, traditional classifiers like SVM and KNN are still clearly outperformed by deep learning algorithms such as YOLOv8, InceptionV3 and MobileNet. Deep learning models are particularly effective at processing large, complex datasets, such as food images, by learning hierarchical features and patterns at various levels of abstraction. This capability allows them to generalize more effectively across a wide range of food types and adapt to different environmental conditions or image qualities, resulting in significantly better performance in terms of both accuracy and speed.

In line with the objective of identifying and analyzing challenges during model training for food image recognition, the project explored different preprocessing techniques required by various models. Each model had distinct preprocessing needs, posing challenges related to consistency and efficiency across all approaches. For instance, models like CNN, SVM, and KNN require the image to be converted to grayscale for feature extraction. Additionally, the images needed to be resized to 224x224 pixels, and the pixel values were normalized to the range [0, 1]. In the case of KNN, each 2D image was flattened into a 1D vector, adding another layer of complexity in terms of processing and memory requirements. On the other hand, the Inception-v3 model requires RGB images (3 color channels). Unlike other models, YOLOv8 automates the data preprocessing step, including image resizing, normalization, and data augmentation, ensuring the preprocessing operations were applied correctly without requiring manual adjustments. By addressing these varying preprocessing needs, we ensured that each model received the appropriate format for optimal performance.

The food image recognition with calorie estimation is implemented by deploying the best model, YOLOv8 into user-friendly Streamlit-based web application. This deployment allows users to upload an image of the food in several formats like jpg, png and webp format. The image can be any class from the 11 food classes that we trained on the model, including Chicken Chop, Chicken Rice, Fish Soup, Kaya Toast, Nasi Lemak, Popiah, Porridge, Prata, Prawn Noodle, Wan Tan Mee and Kuih Lapis. Once the image is uploaded, the model will process and recognize the food items. The recognized food names will then pass to the Nutritionix API to retrieve the corresponding nutritional information. It will display the predicted food and estimated calorie and nutritional information to the user. Users can seamlessly interact with the application, ensuring that the solution is both practical and user-friendly.

5.2. Limitations and Future Works

One of the key limitations of our project is the imbalance in the dataset, where certain classes, particularly “Kuih Lapis,” have significantly more images than others, leading the model to favor these more frequent classes. Additionally, the current images may not fully represent real-world variations, such as differences in plating styles, lighting conditions, and backgrounds. In case, the confusion between similar dishes such as “Chicken Rice”, “Chicken Chop” and “Nasi Lemak” were sometimes classified, highlighting the challenge of fine-grained food recognition. As a result, the model may struggle when classifying images from diverse sources, such as user-submitted photos. Moreover, basic image preprocessing was applied, but more aggressive

image preprocessing could help the model's ability to generalize to unseen images. In the future, we aim to enhance the dataset by expanding the number of classes and increasing the diversity of images to further improve the model's robustness and accuracy in recognizing a wider range of Malaysian dishes.

Reference & Source

Dataset Source References:

Forked Robotflow Repositories:

- <https://universe.roboflow.com/local-bt6la/local-food>
- <https://universe.roboflow.com/malaysia-food/kuih-lapis-amf7c>
- <https://universe.roboflow.com/fyp-ikfwf/wantan-mee>

Dataset Repositories:

- https://universe.roboflow.com/malaysia-food/malaysia_food-r27vd/dataset/2

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Nutritionix API:

- <https://www.nutritionix.com/business/api>

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